PHYSICS

1. What is the dimensional formula of torque?

A) $[LT^{-2}]$ B) $[M L^2T^{-2}]$ C) $[M L^2T^{-1}]$ D) $[M L^{-1}T^{-2}]$

2. The SI unit of surface tension is:

A) N m B) N/m C) J/m D) N/m^2

3. The law of conservation of angular momentum states that the angular momentum of a system remains constant if:

A) The net external force acting on it is zero. B) The net internal force acting on it is zero.
C) The net external torque acting on it is zero. D) The net internal torque acting on it is zero.

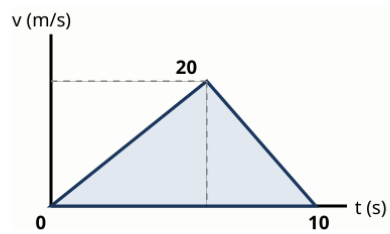
4. A man is inside a lift that is falling freely under gravity. If he drops a coin, the acceleration of the coin with respect to the man is:

A) g B) -g C) Zero D) 2g

5. The moment of inertia of a solid cylinder of mass M and radius R about its own central geometrical axis is:

A) MR^2 B) $1/2\ MR^2$ C) $14\ MR^2$ D) $2/5\ MR^2$

6. The graph below shows the velocity-time curve of a particle. What is the total displacement of the particle in 10 seconds?



A) 50 m B) 100 m
C) 150 m D) 200 m

7. A car travelling at 72 km/h is stopped uniformly in exactly 5 seconds by applying brakes. What is the magnitude of its retardation?

A) $2\ m/s^2$ B) $4\ m/s^2$ C) $8\ m/s^2$ D) $14.4\ m/s^2$

8. If air resistance is neglected, the trajectory of a projectile fired at an angle to the horizontal is strictly a:

A) Circle B) Ellipse C) Parabola D) Straight line

9. The escape velocity of a body from the surface of the Earth depends on the mass of the body (m) as:

A) m B) m^2 C) m^{-1} D) m^0 (Independent of mass)

10. When a satellite revolves very close to the surface of the Earth, its orbital velocity is given by:

A) $\sqrt{gR}$ B) $\sqrt{2gR}$ C) gR D) 2gR

11. The zeroth law of thermodynamics fundamentally leads to the concept of:

A) Internal Energy B) Work C) Temperature D) Entropy

12. At what specific temperature do the Celsius and Fahrenheit temperature scales yield the exact same numerical reading?

A) -40^0 B) 0^0 C) 40^0 D) 100^0

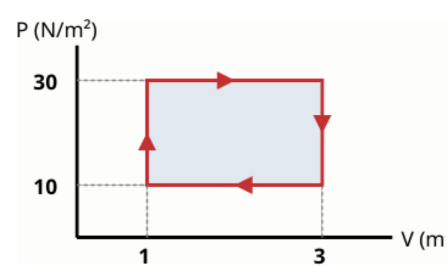
13. In a purely isochoric thermodynamic process, the work done by an ideal gas is:

A) Equal to the heat supplied B) Equal to the change in internal energy C) Maximum D) Zero

14. A perfectly black body is defined as one which:

A) Reflects all radiations incident on it. B) Absorbs all radiations incident on it, regardless of wavelength.
C) Neither absorbs nor emits any radiation. D) Appears completely white at absolute zero.

15. The P-V diagram below represents a cyclic process undergone by an ideal gas. Calculate the total work done by the gas during one complete cycle.



A) 20 J B) 40 J
C) 60 J D) 80 J

16. What is the theoretical efficiency of a Carnot engine operating between a source at 127^0C and a sink at 27^0C'

A) 25% B) 33.3% C) 50% D) 75%

17. Which specific mode of heat transfer does NOT strictly require any material medium for propagation?

A) Conduction B) Convection C) Radiation D) Evaporation

18. The phenomenon of bending of light around the sharp corners of an opaque obstacle is termed:

A) Dispersion B) Diffraction C) Polarization D) Interference

19. In the visible spectrum of light, the refractive index of a typical glass prism is maximum for which color?

A) Red B) Yellow C) Green D) Violet

20. Two thin lenses of power +5 D and -2 D are placed in direct contact. The focal length of the resulting combination is:

A) +33.3 cm B) -33.3 cm C) +10 cm D) -10 cm

21. Why is a convex mirror predominantly used as a rear-view mirror in vehicles?

A) It forms a real, magnified image.

B) It forms a virtual, erect, and diminished image, giving a much wider field of view.

C) It completely eliminates spherical aberration.

D) It absorbs annoying glare from headlights.

22. In Young's double-slit experiment, the fringe width β is directly proportional to:

A) The distance between the two slits B) The frequency of the light used

C) The wavelength of the light used D) The intensity of the light sources

23. If the critical angle for total internal reflection for a given medium against vacuum is 30^0, what is the refractive index of that medium?

A) 1.5 B) 2.0 C) 1.33 D) 2.42

24. The magnifying power of a simple microscope (convex lens) when the final image is formed at the least distance of distinct vision (D) is:

A) D/f B) 1+ (D/f) C) 1-(D/f) D) f/D

25. The defect of vision called myopia (short-sightedness) is clinically corrected by using a:

A) Convex lens B) Concave lens C) Cylindrical lens D) Bifocal lens

26. The reciprocal of electrical resistivity is precisely termed as:

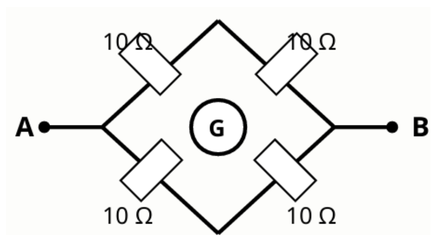
A) Resistance B) Conductance C) Conductivity D) Permittivity

27. Which rule is typically used to definitively find the direction of the magnetic field due to a straight, current-carrying conductor?

A) Fleming's Left-Hand Rule B) Right-Hand Thumb Rule C) Faraday's Rule D) Lenz's Rule

28. When three identically rated resistors, each of resistance R, are connected in parallel, the equivalent resistance of the combination is:

A) 3R B) R/3 C) R D) 3/R



29. Observe the Wheatstone bridge circuit shown below. If the galvanometer (G) shows absolutely no deflection, what is the equivalent resistance across the terminals A and B?

A) 5 Ω B) 10 Ω

C) 20 Ω D) 40 Ω

30. A steady direct current of 5 A flows through a conductive wire for exactly 2 minutes. What is the total electric charge that passed through the wire?

A) 10 C B) 150 C C) 300 C D) 600 C

31. A straight conductor of length 0.5 m carrying a current of 2 A is placed completely perpendicular to a uniform magnetic field of 0.1 T. The magnetic force experienced by the wire is:

A) 0.05 N B) 0.1 N C) 0.2 N D) 1.0 N

32. The soft iron core of a transformer is thoroughly laminated primarily to systematically reduce:

A) Hysteresis losses B) Copper losses C) Eddy current losses D) Flux leakage

33. The SI unit of magnetic dipole moment is:

A) A m B) A m^2 C) T m D) Wb/m^2

34. Magnetic materials that are weakly repelled by an external strong magnet are technically classified as:
A) Ferromagnetic B) Paramagnetic C) Diamagnetic D) Ferrimagnetic

35. The electric field intensity exactly inside a uniformly charged, hollow spherical conducting shell is:
A) Zero B) Constant but non-zero
C) Proportional to distance from the center D) Maximum at the center

36. According to Coulomb's Law, if the distance between two point charges is doubled, the electrostatic force between them mathematically becomes:
A) Doubled B) Halved C) One-fourth D) Four times

37. Two capacitors of capacitance 2 μF and 4 μF are connected strictly in series. Their equivalent combined capacitance is:
A) 6.00 μF B) 1.33 μF C) 0.75 μF D) 8.00 μF

38. The energy stored in a charged capacitor of capacitance C having a potential difference V is given by the formula:
A) cv^2 B) $1/2\ C^2V$ C) $1/2\ CV^2$ D) Q/2C

39. Which elementary particle travels at the speed of light, possesses zero rest mass, and carries zero electrical charge?
A) Electron B) Neutron C) Photon D) Neutrino

40. Cathode rays travelling in a vacuum tube are deflected by:
A) Only electric fields B) Only magnetic fields
C) Both electric and magnetic fields D) Neither electric nor magnetic fields

41. Identify the specific digital logic gate combination represented by the circuit symbol.

A) AND Gate B) OR Gate
C) NAND Gate D) NOR Gate

42. In an n-type semiconductor crystal at room temperature, the minority charge carriers are strictly:
A) Free electrons B) Holes C) Protons D) Positive ions

43. The half-life of a radioactive substance is 5 years. What exact fraction of the original substance will remain perfectly undecayed after 15 years?
A) 1/3 B) 1/4 C) 1/8 D) 1/16

44. Which prominent spectral series of the Hydrogen atom lies entirely in the visible region of the electromagnetic spectrum?
A) Lyman series B) Balmer series C) Paschen series D) Brackett series

45. Isotopes of a given chemical element are defined as atoms possessing:
A) The same number of protons but different number of neutrons.
B) The same number of neutrons but different number of protons.
C) The same mass number but different atomic number.
D) The same number of electrons but different number of protons.

46. According to the Bohr model of the atom, as the principal quantum number (n) consecutively increases, the energy gap between adjacent levels:
A) Increases B) Decreases C) Remains totally constant D) Alternates periodically

47. The work function of a certain photosensitive metal is 2.0 eV. What is the approximate threshold wavelength for photoelectric emission to occur? (Use hc ≈ 12400 eV.Å)
A) 3100 Å B) 6200 Å C) 12400 Å D) 24800 Å

48. The physical process of liberating free electrons from the surface of a metal by intensely heating it is called:
A) Photoelectric emission B) Field emission C) Thermionic emission D) Secondary emission

49. In a standard full-wave bridge rectifier, if the frequency of the input AC supply is 50 Hz, the ripple frequency of the output DC is:
A) 25 Hz B) 50 Hz C) 100 Hz D) 200 Hz

50. Albert Einstein's famous photoelectric equation, $K_{max} = h\nu - \Phi$, is a fundamental manifestation of the law of conservation of:
A) Momentum B) Charge C) Mass D) Energy

CHEMISTRY

51. What is the value of the Bohr radius (a0) for the first orbit of a hydrogen atom?
A) 0.529 Å B) 1.058 Å C) 2.116 Å D) 0.0529 Å
52. According to Graham's Law, the rate of effusion of a gas is inversely proportional to the:
A) Square of its temperature B) Square root of its molar mass
C) Cube root of its volume D) Direct value of its pressure
53. Hess's Law of Constant Heat Summation is fundamentally a direct consequence of the:
A) Zeroth law of thermodynamics B) First law of thermodynamics (Conservation of Energy) C) Second law of thermodynamics D) Third law of thermodynamics
54. All natural artificial radioactive decay processes follow which order of chemical kinetics?
A) Zero order B) First order C) Second order D) Fractional order
55. Which mathematical expression correctly represents Faraday's First Law of Electrolysis?
A) m = Z I / t B) m = Z I t C) Z = m I t D) I = m Z t
56. In a crystalline solid of sodium chloride (NaCl), what is the coordination number of the sodium ion (Na^+)?
A) 4 B) 6 C) 8 D) 12
57. The value of Henry's Law constant (KH) for a given gas dissolved in a liquid:
A) Increases with an increase in temperature B) Decreases with an increase in temperature
C) Remains constant irrespective of temperature D) Is identical for all noble gases
58. What is the SI unit for the cell constant of a conductivity cell?
A) cm B) cm^2 C) m^{-1} D) $\Omega^{-1}\ cm^{-1}$
59. The principle that strictly rules out the concept of definite circular orbits or trajectories for electrons is:
A) Pauli's Exclusion Principle B) Hund's Rule of Maximum Multiplicity
C) Heisenberg's Uncertainty Principle D) Aufbau Principle
60. For any spontaneous irreversible process occurring in an isolated system, the change in entropy (ΔS) must be:
A) Equal to zero B) Less than zero C) Greater than zero D) Dependent on the temperature
61. In a non-ideal binary solution exhibiting a positive deviation from Raoult's Law, how do the intermolecular forces compare?
A) A-B interactions are stronger than A-A and B-B interactions.
B) A-B interactions are weaker than A-A and B-B interactions.
C) A-B interactions are exactly equal to A-A and B-B interactions.
D) A-B interactions do not exist.
62. Why is the boiling point of H_2O significantly higher than that of H_2S, despite Oxygen being lighter than Sulfur?
A) H_2O has a linear geometry. B) H_2O forms extensive intermolecular hydrogen bonds.
C) H_2S is a highly polar molecule. D) Sulfur has available empty d-orbitals.
63. In a standard galvanic cell, which of the following statements accurately describes the anode?
A) It is the positive electrode where reduction occurs. B) It is the negative electrode where reduction occurs. C) It is the positive electrode where oxidation occurs. D) It is the negative electrode where oxidation occurs.
64. If 4 grams of H_2 reacts completely with 32 grams of O_2 to form water, what is the maximum mass of H_2O produced?
A) 18 g B) 34 g C) 36 g D) 72 g
65. Calculate the approximate pH of a 0.01 M solution of acetic acid (CH_3COOH) given its dissociation constant Ka = $1.0 * 10^{-5}$.

A) 2.0 B) 3.5 C) 4.0 D) 5.0

66. The half-life of a specific first-order reaction is 69.3 seconds. What is the rate constant (k) for this reaction?

A) 0.01 s^{-1} B) 0.1 s^{-1} C) 10 s^{-1} D) 100 s^{-1}

67. A standard Daniel cell has E° = 1.10 V. What will be the cell potential if $[Zn^{2+}]$ = 1.0 M and $[Cu^{2+}]$ = 0.01 M at 298 K?

A) 1.10 V B) 1.16 V C) 1.04 V D) 0.98 V

68. Nitrogen has a higher first ionization enthalpy than Oxygen. This anomaly in the periodic table is due to:

A) Smaller atomic size of Nitrogen
B) Greater effective nuclear charge of Nitrogen
C) The extra stability of the exactly half-filled 2p orbital in Nitrogen
D) The presence of a completely filled 2s orbital in Nitrogen

69. In Sulfur Hexafluoride (SF_6), what is the hybridization of the central sulfur atom?

A) sp^3 B) sp^3d C) sp^3d^2 D) d^2sp^3

70. Permanent hardness of water is caused by the presence of soluble:

A) Bicarbonates of Calcium and Magnesium
B) Chlorides and Sulfates of Calcium and Magnesium
C) Carbonates of Sodium and Potassium
D) Phosphates of Calcium

71. When a Barium salt is subjected to a flame test, it imparts which characteristic color to the flame?

A) Crimson red B) Golden yellow C) Apple green D) Brick red

72. The molecular structure of Diborane (B_2H_6) contains a special type of bonding known as:

A) Four-center two-electron bonds B) Three-center two-electron bonds (Banana bonds)
C) Purely ionic bonds D) Coordinate covalent bonds via lone pairs

73. The *'coinage metals'* (Copper, Silver, and Gold) belong to which group of the periodic table?

A) Group 9 B) Group 10 C) Group 11 D) Group 12

74. What is the stable oxidation state of Iron (Fe) in the complex ion $[Fe(CN)_6]^{4-}$?

A) +2 B) +3 C) +4 D) 0

75. Lanthanoid contraction is primarily caused by:

A) Increasing nuclear charge coupled with excellent shielding by 4f electrons.
B) Increasing nuclear charge coupled with very poor shielding by 4f electrons.
C) The decreasing size of the nucleus.
D) Expansion of the 5d orbitals.

76. Why does Nitrogen Trifluoride (NF_3) have a significantly lower dipole moment than Ammonia (NH_3)?

A) NF_3 is a planar molecule, canceling the dipoles.
B) Fluorine is less electronegative than Hydrogen.
C) In NF_3, the orbital dipole of the lone pair opposes the resultant N-F bond dipoles.
D) NF_3 completely lacks a lone pair of electrons.

77. Which of the following oxoacids of Chlorine is the strongest acid?

A) HClO (Hypochlorous acid) B) $HClO_2$ (Chlorous acid) C) $HClO_3$ (Chloric acid) D) $HClO_4$ (Perchloric acid)

78. The diverse colors observed in transition metal complex solutions are most frequently due to:

A) Absorption of light facilitating d-d electron transitions within split d-orbitals.
B) Ligand to Metal Charge Transfer (LMCT) only.
C) The physical state of the solvent.
D) Excitation of electrons from s to p orbitals.

79. According to the Ellingham Diagram, a metal M_1 can successfully reduce the oxide of metal M_2 if:

A) The ΔG vs T curve for M_1 is positioned below the curve for M_2 at that temperature.
B) The ΔG vs T curve for M_1 is positioned above the curve for M_2 at that temperature.
C) Both curves intersect exactly at 0°C.

D) M_2 has a higher melting point than M_1.

80. What is the formal charge on the central oxygen atom in the Ozone (O_3) molecule?

A) -1 B) 0 C) +1 D) +2

81. Calculate the Effective Atomic Number (EAN) of Cobalt (Co, Atomic No: 27) in the complex $[Co(NH_3)_6]^{3+}$

A) 34 B) 35 C) 36 D) 37

82. Which property defines a chelating ligand?

A) It carries a neutral charge.
B) It has a single donor atom.
C) It attaches to the central metal ion through two or more donor atoms to form a ring.
D) It always acts as an oxidizing agent.

83. What is the correct IUPAC name of the compound: CH_3-CH(OH)-CH_2-COOH?

A) 2-Hydroxybutanoic acid B) 3-Hydroxybutanoic acid C) 4-Hydroxybutanoic acid D) Butan-3-ol-1-oic acid

84. How many total structural isomers are possible for an alkane with the molecular formula C_5H_{12}?

A) 2 B) 3 C) 4 D) 5

85. The Wurtz reaction, involving alkyl halides and sodium metal in dry ether, is best utilized for preparing:

A) Symmetrical alkanes containing an even number of carbon atoms.
B) Asymmetrical alkanes containing an odd number of carbon atoms.
C) Alkenes and alkynes exclusively.
D) Aromatic hydrocarbons.

86. The Swarts reaction is a specialized method specifically used for the synthesis of:

A) Alkyl chlorides B) Alkyl bromides C) Alkyl fluorides D) Alkyl iodides

87. Lucas reagent, used to differentiate primary, secondary, and tertiary alcohols, is a mixture of:

A) Concentrated HNO_3 and Anhydrous $ZnCl_2$ B) Concentrated HCl and Anhydrous $ZnCl_2$
C) Concentrated H_2SO_4 and $K_2Cr_2O_7$ D) Dilute HCl and $AlCl_3$

88. Tollens' reagent, which gives a 'silver mirror' test with aldehydes, is chemically:

A) Alkaline potassium permanganate B) Ammoniacal silver nitrate solution
C) Alkaline copper sulfate mixed with Rochelle salt D) Copper(I) chloride dissolved in ammonium hydroxide

89. The Carbylamine test, characterized by the evolution of a foul-smelling gas (an isocyanide), is positive exclusively for:

A) Primary amines (aliphatic and aromatic) B) Secondary amines
C) Tertiary amines D) Amides

90. In maltose, the two D-glucose units are linked together by which specific glycosidic bond?

A) β-1,4 glycosidic linkage B) α-1,4 glycosidic linkage
C) α-1,2 glycosidic linkage D) β-1,2 glycosidic linkage

91. Natural rubber is essentially a linear polymer of which monomer unit?

A) Chloroprene B) Isoprene (2-methyl-1,3-butadiene) C) Neoprene D) Butadiene and Styrene

92. In a bimolecular nucleophilic substitution (SN_2) mechanism, the spatial arrangement of the product relative to the reactant exhibits:

A) Complete retention of configuration B) Racemization
C) Complete inversion of configuration (Walden inversion) D) Isomerization into a structural isomer

93. Arrange the following amines in decreasing order of their basic strength in an aqueous solution: Methylamine, Dimethylamine, Trimethylamine, and Ammonia.

A) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2 > NH_3$ B) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$
C) $CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N > NH_3$ D) $NH_3 > CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$

94. According to Huckel's rule, a cyclic, completely conjugated planar system is aromatic if it contains:

A) 4n π electrons B) (4n+2) π electrons C) (2n+4) π electrons D) Odd number of π electrons

95. Which of the following conditions is an absolute requirement for an aldehyde or ketone to undergo the Aldol Condensation reaction?

A) The presence of a strong oxidizing agent.
B) The presence of an aromatic ring.
C) The presence of at least one α-hydrogen atom.
D) The absence of any α-hydrogen atoms.

96. How does the attachment of an Electron Withdrawing Group (EWG) like $-NO_2$ affect the acidity of a carboxylic acid?

A) It decreases acidity by destabilizing the carboxylate ion.
B) It increases acidity by stabilizing the conjugate base (carboxylate ion) via the -I effect.
C) It has no effect on acidity.
D) It converts the acid into a base.

97. Why is chlorobenzene significantly less reactive towards nucleophilic substitution reactions than chloroethane?

A) Due to the partial double bond character of the C-Cl bond arising from resonance.
B) Due to the sp3 hybridization of the carbon bonded to chlorine.
C) Chlorobenzene is highly soluble in water.
D) Due to the absence of pi electrons in the benzene ring.

98. The reaction of a Grignard reagent (RMgX) with solid Carbon Dioxide (dry ice) followed by acid hydrolysis directly yields a:

A) Primary alcohol
B) Secondary alcohol
C) Ketone
D) Carboxylic acid

99. Complete ozonolysis (O_3 followed by Zn/H_2O) of 2-butene produces:

A) Two molecules of Formaldehyde (Methanal)
B) Two molecules of Acetaldehyde (Ethanal)
C) One molecule of Acetone and one of Formaldehyde
D) Two molecules of Acetic Acid

100. According to Zaitsev's rule, what is the major organic product formed upon the acid-catalyzed dehydration of 2-butanol?

A) 1-Butene
B) 2-Butene
C) 2-Methylpropene
D) Butyne

BOTANY

101. Molisch test is used for the detection of:

A) Carbohydrate
B) protein
C) amino acids
D) fatty acid

102. Sucrose is functionally classified as a non-reducing sugar, whereas maltose and lactose are reducing sugars. What structural feature of the sucrose molecule accounts for this difference?

A) It contains a specialized beta-1,4-glycosidic linkage that is too structurally rigid to break.
B) The glycosidic bond is formed between the anomeric carbon 1 of glucose and the anomeric carbon 2 of fructose, masking both reducing groups.
C) Fructose undergoes immediate intramolecular ring opening, turning into a non-reactive keto group.
D) Sucrose lacks any hexose subunits in its chemical backbone structure.

103. In the micro-anatomy of Gram-negative bacterial cell walls, which unique component provides structural antigenicity and behaves as a potent endotoxin during clinical systemic bacteremia?

A) Teichoic acids anchored to peptidoglycan
B) Lipopolysaccharide (LPS) layer of the outer membrane
C) Ergosterol embedded in the inner plasma membrane
D) Pseudomurein structural protein blocks

104. In the reproductive cycle of the life-form Spirogyra, lateral conjugation differs from scalariform conjugation primarily because lateral conjugation occurs between:

A) Two entirely different filaments positioned parallel to each other.
B) Two adjacent cells of the exact same filament.
C) Male microgametes and female macrogametophytes of distinct populations.
D) Aplanospores formed within distinct specialized zoosporangia.

105. In the classification of true fungi, which group is uniquely characterized by the absolute absence of a flagellated motile stage at any point in its entire life cycle, combined with the production of endogenous sexual spores inside an elongated microscopic cell?

A) Phycomycetes (Chytrids) B) Ascomycetes C) Basidiomycetes D) Deuteromycetes

106. In the morphological anatomy of the liverwort Marchantia, which specialized internal structure is responsible for the expulsion of spores from the mature sporophyte via hygroscopic twisting movements?

A) Pseudo-elaters with spiral thickenings B) True unicellular elaters with spiral bands

C) Peristome teeth rings D) Columella sterile tissue shafts

107. In the life cycle of the fern Dryopteris, the vascular bundles of the mature underground stem (rhizome) are heavily dissected by leaf gaps, presenting a specific anatomical configuration classified as a:

A) Protostele B) Haplostele C) Dictyostele D) Atactostele

108. The leaves of the genus Pinus exhibit prominent xerophytic adaptations to survive in conditions of extreme physiological drought or freezing climates. Which of the following anatomical features represents one of these dedicated adaptations?

A) Thin un-cutinized epidermal layer with raised stomata

B) Sunken stomata placed deeply within pits, combined with a highly thick sclerenchymatous hypodermis

C) Complete absence of any internal transfusion tissue surrounding the vascular strands

D) Massive air cavities (aerenchyma) inside the central mesophyll zone

109. A botany student notes that a specific plant specimen displays zygomorphic flowers with a papilionaceous corolla (standard, wings, and keel petals), ten diadelphous stamens configured as (9)+1, and a monocarpellary unilocular superior ovary with marginal placentation. This plant belongs to the family:

A) Brassicaceae B) Solanaceae C) Fabaceae D) Liliaceae

110. In virus biology, the lysogenic cycle of a temperate bacteriophage differs fundamentally from the lytic cycle because in the lysogenic cycle:

A) The viral capsule directly lyses the host plasma membrane from the outside.

B) The viral genome integrates cleanly into the host bacterial chromosome, replicating silently as a prophage without killing the host cell.

C) Host protein synthesis machinery is destroyed within minutes of viral DNA injection.

D) The virus utilizes specialized reverse transcriptase to synthesize viral ribozymes.

111. The traditional medicinal plant Rauwolfia serpentina (Sarpagandha) is highly prized in modern pharmacology. Its active secondary metabolite, reserpine, is purified and used globally in clinical medicine primarily to treat which human pathophysiological state?

A) Acute respiratory hypotension

B) Chronic systemic hypertension and neuropsychiatric agitation

C) Acute bacterial septicemia

D) Severe insulin-dependent diabetes mellitus

112. In the global nitrogen cycle, which specific genus of soil bacteria is biochemically responsible for the oxidative conversion of toxic nitrites (NO^{2-}) into stable nitrates (NO^{3-}) during the process of nitrification?

A) Nitrosomonas B) Nitrobacter C) Rhizobium D) Azotobacter

113. During plant ecological succession originating within an open aquatic environment (Hydrosere), which specific community of plants forms the transitional 'reed-swamp stage'?

A) Microscopic free-swimming phytoplanktons like diatoms

B) Submerged rooted macrophytes like Hydrilla and Vallisneria

C) Amphibious rooted plants like Typha, Phragmites, and Sagittaria

D) Terrestrial woody tree species like Alnus and Salix

114. At a molecular level, how do Chlorofluorocarbons (CFCs) drive the catastrophic catalytic destruction of the stratospheric ozone layer?

A) CFC molecules physically dissolve ozone gas via non-polar solvent interactions.

B) UV radiation homolytically cleaves CFCs to release highly reactive free chlorine radicals, each capable of breaking down thousands of ozone (O_3) molecules into oxygen (O_2).

C) CFCs bind to oxygen gas, completely blocking the photochemical synthesis of ozone by sunlight.

D) They heavily increase greenhouse trapping, warming the stratosphere until ozone molecules thermally decompose.

115. A high-altitude forest survey across the sub-alpine bioclimatic zones of Nepal (between 3,000 to 3,800 meters) reveals a dense forest canopy dominated by trees with peeling, papery bark, coupled with stunted evergreen shrubs. This native ecosystem is classified as a:

A) Tropical Deciduous Shorea robusta forest ` B) Sub-tropical Pinus roxburghii forest
C) Betula utilis (Bhojpatra) and Rhododendron forest D) Pure Oak (Quercus) temperate cloud forest

116. According to the fluid mosaic model of biomembranes, which type of lipid-protein movement is exceptionally rare and requires high metabolic energy inputs due to the hydrophilic polar heads crossing the hydrophobic core?

A) Lateral diffusion of proteins within the monolayer plane
B) Rotational spinning of integral proteins along their long axis
C) Transverse 'flip-flop' movement of lipids from one leaflet to the opposing leaflet
D) Flexing movement of non-polar fatty acid hydrocarbon tails

117. Mitochondria and chloroplasts are classified as semi-autonomous organelles. Which of the following structural traits provides direct molecular evidence supporting their endosymbiotic prokaryotic origin?

A) The presence of linear eukaryotic chromosomes bundled with histone proteins.
B) The absolute presence of an 80S translation ribosome complex inside their matrix.
C) The presence of a circular, naked DNA genome coupled with prokaryotic-like 70S ribosomes.
D) Their ability to synthesize all required structural proteins entirely independent of nuclear genes.

118. In the structural architecture of a eukaryotic chromosome, the highly specialized regions located at the absolute terminal ends that protect the coding DNA from degradation and shorten with each cycle of replication are termed:

A) Centromeres B) Kinetochores C) Telomeres D) Satellites

119. longest phase in cell cycle is

A) prophase B) interphase C) metaphase D) anaphase

120. In experimental cytogenetics, onion root tips are treated with the chemical alkaloid Colchicine to perform chromosome counting. What is the molecular mechanism of action of colchicine, and at which mitotic sub-stage are the cells arrested?

A) It stabilizes actin filaments, arresting cells in Anaphase.
B) It binds to tubulin dimers, preventing spindle microtubule polymerization and arresting cells completely in Metaphase.
C) It breaks down nuclear envelopes prematurely, trapping cells in Prophase.
D) It prevents cytokinesis by disrupting cell plate formation during Telophase.

121. According to the structural parameters of standard B-DNA established by Watson and Crick, a single complete helical turn spans a distance of 3.4 nm and contains exactly how many base pairs?

A) 12 base pairs B) 10 base pairs C) 20 base pairs D) 34 base pairs

122. Okazaki fragments are present in which strand?

A) Leading B) lagging C) both D) neither a nor b

123. Which of the following human genetic syndromes is correctly classified as an allosomal (sex chromosome) aneuploidy caused by meiotic non-disjunction, resulting in a 47,XXY karyotype?

A) Down's syndrome B) Turner's syndrome C) Klinefelter's syndrome D) Edward's syndrome

124. In genetic crosses, a testcross involving an F1 dihybrid individual yields an unexpected phenotypic ratio of 7:1:1:7 instead of the classic Mendelian independent assortment ratio of 1:1:1:1. This outcome indicates that the two genes under investigation are:

A) Located on completely different, unlinked non-homologous chromosomes.
B) Syntenic and tightly linked together in the coupling (cis) configuration.
C) Exhibiting a highly complex form of duplicate dominant epistasis.
D) Inherited strictly via maternal cytoplasmic inheritance loops.

125. A double-stranded DNA fragment isolated from a plant pathogen is biochemically analyzed and found to contain 32% Adenine bases. What is the calculated percentage of Cytosine bases present in this DNA strand?

A) 32% B) 68% C) 18% D) 16%

126. In an experimental agricultural plot, researchers crossed a diploid crop plant (2n = 14) with a tetraploid variant of the same species (4n = 28). What is the expected somatoplastic chromosome number and the fertility status of the resulting F1 triploid offspring?

A) 2n = 21; completely fertile due to balanced binary mitotic divisions

B) 2n = 42; highly sterile due to structural chromosome breakage

C) 2n = 21; highly sterile because the odd ploidy level prevents normal homologous chromosome pairing during Meiosis I

D) 2n = 28; fully fertile autopolyploid variant lines

127. In the stem anatomy of several families like Cucurbitaceae and Solanaceae, the vascular bundles contain two distinct strands of phloem positioned on both the outer and inner margins of the central xylem core. This specialized bundle configuration is termed:

A) Collateral open bundle B) Bicollateral bundle

C) Concentric amphivasal bundle D) Radial closed bundle

128. When comparing the transverse section of a monocot root with a dicot root under a microscope, which of the following anatomical markers can be used to explicitly identify the specimen as a monocot root?

A) Presence of a small, poorly developed or completely absent central pith.

B) A diarch to hexarch xylem configuration with an active secondary cambium ring.

C) A polyarch xylem configuration with more than six distinct xylem bundles, surrounding a large, well-developed central parenchymatous pith.

D) Complete absence of an organized endodermal layer with Casparian strips.

129. In a forensic investigation, a section of woody tissue from an ancient structural beam is analyzed. The wood shows distinct growth rings, possesses wide vessels with large lumens, and lacks any thick fiber clusters. This specific section represents wood that was formed during which season, and what is it termed?

A) Autumn season; Autumn wood (Late wood) B) Spring season; Spring wood (Early wood)

C) Winter dormancy; Heartwood core layers D) Continuous dry seasons; Tyloses filled sapwood

130. Water potential (Yw) is a fundamental parameter governing water kinetics in plant systems. According to thermodynamic conventions, what is the exact water potential value assigned to pure water under standard temperature and atmospheric pressure?

A) -10 bars B) +1.0 MPa C) Zero D) Dependent on cell wall elasticity parameters

131. During the light-dependent reactions of photosynthesis, the primary purpose of the non cyclic electron transport chain (the Z-scheme) is to synthesize which energy-rich compounds for the Calvin cycle?

A) ATP and NAD^+ B) ATP and NADPH + H^+

C) Glucose and Oxygen gas D) $FADH_2$ and Rubisco enzymes

132. Photorespiration (C_2 cycle) reduces the overall photosynthetic efficiency of C_3 plants. This metabolic pathway is initiated when the primary carbon-fixing enzyme Rubisco oxygenates Ribulose-1,5-bisphosphate (RuBP) instead of carboxylation, directly yielding:

A) Two molecules of 3-Phosphoglyceric acid (PGA)

B) One molecule of 3-PGA and one molecule of 2-Phosphoglycolate

C) One molecule of Oxaloacetic acid (OAA)

D) Two molecules of Malic acid

133. In the mitochondrial matrix, the Krebs cycle generates reducing equivalents (NADH, FADH2). However, there is exactly one enzymatic step that produces a molecule of high-energy nucleoside triphosphate via direct substrate-level phosphorylation. This step occurs during the conversion of:

A) Isocitrate into alpha-Ketoglutarate B) Succinyl-CoA into Succinate

C) Succinate into Fumarate via succinate dehydrogenase D) Malate into Oxaloacetate

134. A plant physiology laboratory monitors three adjacent parenchymal cells: Cell X, Cell Y, and Cell Z. Cell X has an Osmotic Pressure (OP) of 10 atm and a Turgor Pressure (TP) of 6 atm. Cell Y has an OP of 12 atm and a TP of 4 atm. Cell Z has an OP of 8 atm and a TP of 5 atm. What is the calculated Diffusion Pressure Deficit (DPD) for each cell, and what will be the direction of net water movement among them?

A) $DPD_X = 16$, $DPD_Y = 16$, $DPD_Z = 13$; water will move from Cell Y to Cell Z.
B) $DPD_X = 4$ atm, $DPD_Y = 8$ atm, $DPD_Z = 3$ atm; water will move from Cell Z and Cell X into Cell Y.
C) $DPD_X = 6$, $DPD_Y = 4$, $DPD_Z = 5$; water flows uniformly in a closed circular loop.
D) $DPD_X = 0$, $DPD_Y = 2$, $DPD_Z = 1$; water will flow strictly from Cell Y into Cell X.

135. A commercial horticulturist wishes to spray a field of dwarf pea crops to induce rapid stem elongation, break the seed dormancy of a batch of light-sensitive seeds, and promote the production of amylase enzymes in malting barley. Which plant growth regulator should they select?

A) Abscisic acid (ABA) B) Gibberellic acid (GA_3)
C) Zeatin (Cytokinin) D) Indole-3-acetic acid (Auxin)

136. In the typical developmental monosporic pathway (Polygonum type) of an angiosperm ovule, a single functional megaspore undergoes three sequential mitotic nuclear divisions. The resulting mature, unfertilized embryo sac exhibits which cellular and nuclear configuration?

A) 8 cells and 8 nuclei B) 7 cells and 7 nuclei
C) 7 cells and 8 nuclei D) 3 cells and 8 nuclei

137. Most common type of ovule is

A) amphitropous B) orthrotropous C) anatropous D) camphylotropous

138. In traditional crop improvement and plant breeding practices, the surgical removal of immature anthers from a bisexual flower bud prior to their dehiscence is termed:

A) Bagging B) Emasculation C) Clonal selection D) Heterosis stabilization

139. In plant genetic engineering, the soil bacterium Agrobacterium tumefaciens is widely exploited as a natural vector for gene transfer. What specific segment of its tumor-inducing (Ti) plasmid is transferred and integrated into the host plant genome?

A) vir genes (virulence region) B) T-DNA (Transfer DNA) region
C) OriC (Origin of replication) D) Opine catabolism genes

140. A biotechnology research team aims to create a somatic hybrid plant with frost resistance and high starch yield. They isolate leaf mesophyll cells from a wild potato species and a cultivated potato variety. To successfully achieve protoplast fusion (somatic hybridization), which chemical agent should they use?

A) Colchicine solution B) Polyethylene Glycol (PEG) and high Ca^{2+} concentration
C) Cellulase and pectinase cocktail mixtures D) Agrobacterium liquid suspensions

ZOOLOGY

141. In molecular phylogenetics, amino acid sequence comparison of conserved proteins serves as a reliable evolutionary clock. When analyzing Cytochrome c across different taxa, which of the following molecules shows the closest structural alignment and fewest sequence variations relative to human Cytochrome c?

A) Rhesus Monkey B) Chimpanzee C) Baker's Yeast D) Gray Whale

142. August Weismann elegantly dismantled Jean-Baptiste Lamarck's theory of the 'Inheritance of Acquired Characters' by cutting off the tails of mice for over 20 consecutive generations. Which of the following fundamental insights represents Weismann's core theoretical justification?

A) Phenotypic modifications can selectively induce localized reverse-transcription in somatic lines.
B) Somatoplasm variations do not influence the genetic architecture of germplasm cells within the gonads.
C) Acquired variations are inherently dominant over ancestral germinal traits.
D) Environmental pressures selectively preserve mutations exclusively during embryogenesis.

143. In a remote, isolated human population in a mountainous district of Nepal, a rare autosomal recessive metabolic disorder exhibits a phenotypic frequency of 1 in 10,000 individuals. Assuming this

population satisfies all criteria for ideal Hardy-Weinberg equilibrium, what is the calculated frequency of healthy, non-symptomatic heterozygous carriers within this population?

A) 0.01 B) 0.02 C) 0.18 D) 0.0198

144. In the architectural micro-anatomy of a cnidarian cnidocyte, the structural mechanoreceptor trigger that projects outward to detect passing prey items and induce the explosive discharge of the nematocyst is termed the:

A) Operculum B) Cnidocil C) Refractile Lasso D) Stylet arch

145. Respiratory organs across Phylum Arthropoda vary extensively due to adaptive radiation across aquatic and terrestrial habitats. Which of the following arthropod groups is uniquely paired with its correct dedicated respiratory structure?

A) Scorpions - Tracheal tubes B) Prawns (Palaemon) - Book lungs
C) King Crab (Limulus) - Book gills D) Centipedes - Dermal branchiae

146. True coelomates are structurally classified as either schizocoelous or enterocoelous based on their embryonic development patterns. How does a schizocoelom differ fundamentally from an enterocoelom during morphogenesis?

A) A schizocoelom forms by the direct conversion of the persistent embryonic blastocoel.
B) A schizocoelom arises from the physical splitting of solid bilateral bands of embryonic mesoderm.
C) A schizocoelom develops through the outward pouching of the embryonic archenteron wall.
D) A schizocoelom is entirely devoid of any supportive internal mesodermal peritoneum lining.

147. A marine zoological expedition harvests an unsegmented, worm-like benthic burrowing animal. Microscopic analysis reveals the presence of a muscular proboscis, pharyngeal gill slits, and an anterior buccal diverticulum extending into the proboscis cavity, but no true rigid dorsal notochord. This specimen should be classified as:

A) Balanoglossus (Hemichordata) B) Amphioxus (Cephalochordata)
C) Herdmania (Urochordata) D) Petromyzon (Cyclostomata)

148. Microvilli and cilia represent distinct specialized modifications of the epithelial cell apical surface. Which of the following options correctly identifies the unique location and underlying structural feature of 'stereocilia' within the male reproductive anatomy?

A) Lining of the fallopian tubes; composed of motile microtubular axonemes
B) Lining of the epididymis; composed of non-motile, elongated actin filament bundles
C) Lining of the trachea; driven by ATP-dependent dynein motor arms
D) Lining of the seminiferous tubules; specialized for rapid phagocytosis of flagella

149. In the ultrastructural organization of skeletal muscle contraction fibers, a structural feature known as a 'triad' plays a fundamental role in excitation-contraction coupling. A muscle triad is composed of:

A) One thick myosin filament, two thin actin filaments, and an intercalated disc
B) One transverse (T) tubule flanked symmetrically by two terminal cisternae of the sarcoplasmic reticulum
C) Three overlapping actin filaments anchored directly onto a structural Z-line
D) Two troponin complexes attached to a single elongated tropomyosin strand

150. In connective tissue biology, collagen fibers provide exceptional tensile strength to matrices. A patient suffering from a severe systemic deficiency of Ascorbic Acid (Vitamin C) exhibits fragile blood vessels and bleeding gums because Vitamin C is a necessary cofactor for:

A) The initial cross-linking of elastic fibers via desmosine bridges
B) The enzymatic hydroxylation of proline and lysine residues during procollagen synthesis
C) The extracellular cleavage of tropocollagen non-helical terminal peptides
D) The polymerization of chondroitin sulfate within ground substance matrices

151. A postsynaptic somatic motor neuron receives hundreds of localized electrical inputs simultaneously. At a given millisecond, it receives 50 separate Excitatory Postsynaptic Potentials (EPSPs) of +1.5 mV each and 30 separate Inhibitory Postsynaptic Potentials (IPSPs) of -2.0 mV each. If the cell's baseline resting membrane potential is -70 mV and its threshold to fire is -55 mV, which of the following events will occur?

A) The inputs will cancel out perfectly, and the potential will remain at -70 mV.

B) Spatial summation will drive the membrane potential to -55 mV, triggering an action potential at the axon hillock.
C) Temporal summation will hyperpolarize the neural membrane to -85 mV, entering a refractory block.
D) The membrane will depolarize to -60 mV, failing to reach threshold, thus no action potential will fire.

152. In the high-velocity invasion mechanism of the Plasmodium vivax merozoite into a human erythrocyte, which intracellular structural organelle complex secretes specialized proteolytic enzymes to directly create a vacuole for the parasite's entry?
A) Contractile vacuole system
B) The Apical Complex (Rhoptries and Micronemes)
C) Schüffner's granular reticulum
D) Macronuclear chromatin rings

153. In the central nervous architecture of the earthworm Pheretima posthuma, the nerve ring surrounding the pharynx is constructed from pairs of ganglia. The supra-pharyngeal ganglia, acting as a primitive brain, are physically situated in which specific body segment?
A) 1st segment B) 3rd segment C) 7th segment D) 9th segment

154. In the coelom-related reproductive biology of male frogs (Rana tigrina), a small organ known as 'Bidder's Organ' is located at the anterior pole of each testis. Structurally and evolutionarily, Bidder's Organ represents:
A) A vestigial adrenal gland that regulates sodium retention during hibernation.
B) A specialized endocrine structure that directly synthesizes testosterone boosters.
C) An underdeveloped, vestigial ovary that can mature if the testes are surgically removed.
D) A lymphoid filtering station that screens spermatozoa for motility defects.

155. During the reproductive behavior of Pheretima posthuma, cross-fertilization occurs via mutual exchange of sperm. Earthworms have spermathecae that act as specialized storage chambers. What is the exact segment location of these spermathecae, and what do they store?
A) Segments 17 and 19; they store their own autologous sperm prior to mating.
B) Intersegmental apertures of segments 5/6, 6/7, 7/8, and 8/9; they store foreign allogenic sperm received from a partner.
C) Internal walls of segment 14; they store unfertilized mature ova.
D) Coelomic spaces of segments 10 and 11; they function as seminal vesicles.

156. In the circulatory physiology of the frog (Rana), the single ventricle pumps blood into the conus arteriosus. Inside the conus, a longitudinal spiral fold or valve plays a critical role. What is its main physiological function?
A) It acts as an absolute mechanical gate that blocks any blood from reaching the liver tissue.
B) It helps direct deoxygenated blood preferentially into the pulmocutaneous arches and oxygenated blood into the carotid and systemic arches.
C) It serves as a contractile pacemaker that initiates the second stage of the heartbeat.
D) It filters out old, damaged erythrocytes prior to arterial distribution.

157. A soil biology student analyzes the chemical composition of earthworm castings (excreta) and notices that despite eating highly acidic decaying leaf litter and humic soils, the output castings maintain a near-neutral pH (~7.0). This acid-neutralizing capability is directly attributed to the physiological activity of which specific structural organs?
A) Typhlosole intestinal ridges B) Pharyngeal salivary tufts
C) Calciferous glands of the esophagus D) Intestinal gizzard matching filters

158. In the intestinal digestion of carbohydrates, the final breakdown occurs at the microvillar brush border membrane of enterocytes. Which of the following disaccharidases is correctly matched with its specific monosaccharide cleavage products?
A) Maltase - One molecule of Glucose + One molecule of Galactose
B) Lactase - Two molecules of Glucose
C) Sucrase - One molecule of Glucose + One molecule of Fructose
D) Isomaltase - Two molecules of Galactose

159. The transport of carbon dioxide (CO_2) in blood involves a physiological process known as the Haldane Effect. The Haldane Effect states that:

A) High oxygen partial pressure (pO_2) in the lungs promotes the unloading of carbon dioxide from hemoglobin.
B) High carbon dioxide concentrations cause hemoglobin to release its bound oxygen molecules in tissues.
C) Active skeletal muscles promote the rapid synthesis of 2,3-bisphosphoglycerate within erythrocytes.
D) Carbonic acid dissociation is entirely dependent on systemic blood pressure vectors.

160. In the human fetus, the circulatory system bypasses the non-functional, fluid-filled lungs via specialized pathways. The temporary vascular channel that directly connects the pulmonary trunk to the descending aorta is termed the:

A) Foramen ovale B) Ductus venosus C) Ductus arteriosus D) Fossa ovalis

161. The nephron handles water and solutes via differential transport properties across its segments. Which structural segment of the renal tubule is characterized by an absolute structural impermeability to water molecules combined with active transport of sodium (Na^+) and chloride (Cl^-) ions into the interstitium?

A) Thin descending limb of the Loop of Henle B) Proximal Convoluted Tubule (PCT)
C) Thick ascending limb of the Loop of Henle D) Medullary Collecting Duct

162. Human cortical localization associates specific cognitive and motor tasks with dedicated structural areas. A patient who suffers an ischemic stroke involving the left frontal lobe immediately superior to the lateral sulcus loses the ability to physically produce spoken words, though they fully understand written and spoken language. This injury is located in:

A) Wernicke's Area B) Broca's Area
C) Precentral Somatosensory Gyrus D) Angular Corpus Gyrus

163. Photoreception inside retinal rod cells involves a highly stereotypic photochemical cycle. Upon absorption of a single photon of light, the prosthetic visual pigment rhodopsin undergoes immediate activation due to which specific chemical change?

A) The phosphorylation of opsin by active tyrosine kinases
B) The photo-isomerization of 11-cis-retinal into all-trans-retinal
C) The enzymatic conversion of Vitamin A into beta-carotene
D) The open channel influx of cyclic GMP molecules into the outer segment

164. The synthesis of thyroid hormones (T_3 and T_4) by thyroid follicular cells is a multi-step biochemical process. The initial covalent binding of iodine atoms onto tyrosine amino acid residues occurs within which specific site?

A) The rough endoplasmic reticulum of the follicular cell
B) Inside the nuclear matrix of the parafollicular C-cells
C) Extracellularly within the colloid fluid of the follicular lumen
D) The inner mitochondrial membrane matrix

165. During human fertilization, the mammalian spermatozoon must penetrate the protective extracellular coverings of the secondary oocyte. Which specialized enzyme contained within the sperm acrosome is responsible for dissolving the cellular intercellular cement of the corona radiata?

A) Acrosin B) Hyaluronidase C) Cathepsin G D) Acid Phosphatase

166. While hydrophilic nutrient building blocks (monosaccharides and amino acids) are absorbed directly into the intestinal blood capillaries, the end products of lipid digestion follow a different path. Which option correctly explains why lipids enter the lacteals?

A) Large triglycerides are too highly charged to traverse the endothelial cell wall of blood vessels.
B) Inside enterocytes, lipids are re-esterified into triglycerides and packaged into massive lipoprotein spheres called chylomicrons, which are too large to pass through blood capillary pores.
C) Lacteals contain specialized active transport active pumping proteins specific only for free glycerol.
D) Blood capillary endothelial membranes possess enzymes that completely destroy any entering fatty acid chain.

167. When a low-altitude resident travels rapidly to a high-altitude base camp in the Himalayas (>4,000 meters), their body undergoes physiological respiratory compensation to adapt to the lower partial

pressure of oxygen. Within 48 hours, which of the following erythrocyte modifications shifts the oxygen-hemoglobin curve to facilitate oxygen unloading to hypoxia-threatened tissues?
A) A heavy decrease in internal cellular pH within the RBC cytoplasm.
B) Accelerated synthesis of 2,3-bisphosphoglycerate (2,3-BPG), shifting the dissociation curve to the right.
C) The down-regulation of alpha-globin chains to favor embryonic hemoglobin variants.
D) Complete inhibition of carbonic anhydrase to arrest chloride shift dynamics.

168. The Baroreceptor Reflex loop is a vital homeostatic feedback mechanism regulating systemic blood pressure. If an intravenous injection of epinephrine causes a sudden, massive acute spike in blood pressure, how does the baroreceptor reflex respond to restore homeostasis?
A) Decreased firing of baroreceptors, leading to increased sympathetic cardiac accelerator drive.
B) Increased firing of glossopharyngeal and vagal nerve afferents to the medulla, causing increased parasympathetic vagal discharge to slow the heart rate.
C) Selective localized vasoconstriction of the renal cortical vascular beds via aldosterone.
D) Sudden systemic activation of the renin-angiotensin-aldosterone axis to drop vessel compliance.

169. Atrial Natriuretic Peptide (ANP) is a cardiac hormone synthesized and secreted by atrial myocytes in response to high blood volume. Which statement describes the precise mechanism by which ANP lowers blood volume and systemic blood pressure?
A) It acts on the hypothalamus to trigger massive thirst and release antidiuretic hormone.
B) It causes vasoconstriction of afferent arterioles in the kidney to reduce filtration.
C) It dilates afferent renal arterioles, increases GFR, and directly inhibits the secretion of Renin and Aldosterone, promoting sodium and water excretion.
D) It accelerates the active transport reabsorption of sodium ions in the collecting duct.

170. Water-soluble peptide hormones cannot cross the phospholipid membrane of target cells. When a hormone like Epinephrine binds to a beta-1 adrenergic receptor on a cardiac cell, it activates an intracellular cascade. Which sequence accurately traces this G-protein coupled receptor second messenger pathway?
A) Hormone → Activation of Phospholipase C → Production of IP3 and DAG → Calcium influx
B) Hormone → G-protein activation → Stimulation of Adenylate Cyclase → Conversion of ATP to cyclic AMP (cAMP) → Activation of Protein Kinase A
C) Hormone → Direct binding to nuclear chromatin → Activation of RNA polymerase lines
D) Hormone → Tyrosine kinase auto-phosphorylation → Activation of GLUT-4 glucose transporters

171. A patient with severe chronic emphysema presents to an emergency department in Kathmandu with profound respiratory distress. Arterial Blood Gas (ABG) analysis reveals a blood pH of 7.21, a partial pressure of carbon dioxide (pCO_2) of 68 mmHg (normal: 35-45 mmHg), and a bicarbonate (HCO^{3-}) concentration of 28 mEq/L (normal: 22-26 mEq/L). How should this metabolic status be diagnosed?
A) Acute Respiratory Alkalosis with full renal clearance
B) Uncompensated Metabolic Acidosis due to lactic acid retention
C) Respiratory Acidosis with partial metabolic compensation
D) Fully compensated Metabolic Alkalosis

172. A 24-year-old patient who survived a severe motorcycle accident involving a head injury presents with extreme polyuria (passing over 12 Liters of dilute urine per day) and intense polydipsia (excessive thirst). Lab tests reveal urine specific gravity is near that of water (1.002), and urine glucose testing is completely negative. This pathophysiological condition is caused by a structural lesion damaging which endocrine pathway?
A) Autoimmune destruction of the beta-islet cells of the pancreas
B) Fracture of the sella turcica disrupting the supraoptic nuclei of the hypothalamus or posterior pituitary, halting ADH release
C) Hypersecretion of aldosterone from a cortical adenoma
D) Direct nephrogenic resistance to vasopressin caused by hypercalcemia

173. Corynebacterium diphtheriae is a dangerous respiratory pathogen. At a molecular level, the secreted Diphtheria exotoxin causes cell death and tissue necrosis within the human pharynx by directly neutralizing which component of host cellular machinery?
A) It degrades the phospholipids of the nuclear membrane via lipase activity.
B) It catalyzes the ADP-ribose inactivation of Elongation Factor 2 (EF-2), fully arresting host ribosomal protein synthesis.
C) It binds to hemoglobin to prevent oxygen binding.
D) It selectively cleaves the host genomic DNA at palindromic repeat sites.
174. Secretory Immunoglobulin A (IgA) acts as the primary immunological gatekeeper across human mucosal membranes. During its transport across mucosal epithelial cells into secretions (transcytosis), which specific structural component is added to prevent enzymatic proteolytic degradation within the gut lumen?
A) A pentameric J-chain linkage loop
B) The Secretory Component derived from the epithelial polymeric immunoglobulin receptor
C) An IgE Fc-receptor binding domain
D) A specialized disulfide-rich carbohydrate hinge region
175. The Hepatitis B Virus (HBV) is a highly infectious pathogen targeting human hepatocytes. Structurally, it is classified as a Hepadnavirus possessing a unique replication strategy. How does HBV replicate its genetic material inside host cells?
A) Its single-stranded RNA genome serves directly as mRNA for translation without nuclear steps.
B) It possesses a partially double-stranded DNA genome that uses a viral reverse transcriptase enzyme to transcribe its genomic DNA from an RNA pre-genome intermediate.
C) It completely integrates into host autosomes and replicates purely via standard host DNA Polymerase alpha.
D) It utilizes a specialized RNA-dependent RNA polymerase to assemble negative-strand ribo-nucleic loops.
176. A healthcare worker experiences an accidental needle-stick injury while managing an HIV-positive patient. To screen the worker immediately for potential early infection within the first 14 days post exposure (the standard immunological window period), which diagnostic assay provides the highest sensitivity and specificity to confirm or rule out infection?
A) Western Blot assay tracking anti-gp120 IgG antibodies
B) Standard indirect ELISA screening for human anti-HIV antibodies
C) Quantitative Real-Time PCR (RT-PCR) tracking viral HIV-1 RNA copy numbers
D) Ouchterlony double immunodiffusion tracking p24 precipitation lines
177. Transgenic animal technology utilizes mammalian expressions to harvest therapeutic human pharmaceuticals, a process termed 'pharming'. In the classic historical development of transgenic sheep lines (such as *'Tracy'*), the expression of which human protein was engineered into the milk to treat human emphysema?
A) Human Insulin B) Alpha-1-Antitrypsin C) Tissue Plasminogen Activator (tPA) D) Factor VIII
178. Organ transplantation rejection can manifest via different temporal and immunological pathways. A hyperacute graft rejection reaction, occurring within minutes to hours after a kidney transplant vascularization is completed, is structurally driven by:
A) De novo activation of host naive cytotoxic CD8+ T-lymphocytes over several weeks.
B) Pre-existing circulating host antibodies (such as anti-ABO or anti-MHC Class I) binding to graft vascular endothelium, instantly triggering the complement cascade and widespread thrombosis.
C) Delayed-type hypersensitivity macrophages blocking the renal papillae lines.
D) Structural incompatibility between host and donor mitochondrial DNA rings.
179. Birds (Class Aves) exhibit extensive anatomical and physiological modifications to minimize weight for efficient volant (flight) adaptation. Which of the following reproductive features represents a unique, dedicated weight-reduction modification found in the majority of modern female birds?
A) Total absence of shell-secreting uterine tissue.
B) Complete anatomical regression of the right ovary and right oviduct, leaving only the left functional side.

C) Shifting of embryonic incubation completely to external nests managed by males.
D) Continuous excretion of nitrogenous wastes as heavy liquid urea solution pools.

180. In accordance with international conservation policies under the IUCN and the Ramsar Convention, the Government of Nepal manages several highly protected areas. The Koshi Tappu Wildlife Reserve was specifically designated as Nepal's first international Ramsar Site primarily because it serves as a critical, globally significant habitat for:
A) High-altitude breeding pairs of Snow Leopards .
B) Migratory waterfowl and the last surviving native population of the Wild Water Buffalo
C) Relict populations of the alpine Himalayan Newt (Tylototriton verrucosus).
D) Endemic deep-water populations of Blind Ganges River Dolphins inside rapid gorges.

MAT

181. Statements: Only a few Rivers are Lakes, All Lakes are Oceans, No Ocean is a Sea.
Conclusions: I. All Rivers being Oceans is a possibility.
II. Some Lakes are not Seas.
A) Only Conclusion I follows B) Only Conclusion II follows
C) Both Conclusions I and II follow D) Neither Conclusion I nor II follows1

182. What does the code 'pa' stand for in a certain code language?
Statement I: In that code language, 'pa ta ma' means 'water is hot'.
Statement II: In that code language, 'na pa da' means 'cold water flows'.
A) Statement I alone is sufficient to answer the question.
B) Statement II alone is sufficient to answer the question.
C) Both statements I and II together are required to answer the question.
D) Both statements even together are not sufficient to answer.

183. Statement: "To significantly improve the city's air quality, the government has immediately banned all 15-year-old diesel vehicles."
Assumptions: I. Diesel vehicles are the only source of air pollution in the city.
II. 15-year-old diesel vehicles contribute significantly to the air pollution.
A) Only Assumption I is implicit B) Only Assumption II is implicit
C) Both Assumptions I and II are implicit D) Neither Assumption I nor II is implicit

184. Let 'P $ Q' mean 'P is not smaller than Q'.
Let 'P @ Q' mean 'P is neither smaller than nor equal to Q'.
Let 'P # Q' mean 'P is neither greater than nor equal to Q'.
Given Statements: A @ B, B $ C, C # D,
Conclusions: I. A @ C
II. D @ B
A) Only Conclusion I follows B) Only Conclusion II follows
C) Both Conclusions I and II follow D) Neither Conclusion I nor II follows

185. If the word 'NEPAL' is coded as 'MFOZK', how will the word 'KOSHI' be coded in that same language?
A) LNTIJ B) JPRIH C) JPRHJ D) LPTGI

186. A clock gains exactly 10 minutes every 24 hours. It is set right at 8:00 AM on Monday. What will be the true time when the defective clock indicates 6:00 PM on the following Wednesday?
A) 5:00 PM Wednesday B) 5:24 PM Wednesday C) 5:36 PM Wednesday D) 6:10 PM Wednesday

187. Find the wrong term in the following number series: 10, 15, 22, 33, 46, 65, 82
A) 33 B) 46 C) 65 D) 82

188. Advanced Missing Character: Find the missing number in the grid matrix:

18	**24**	**21**
12	**14**	**13**
27	**?**	**33**

A) 29 B) 35 C) 39 D) 41

189. Quantity A: The total number of ways to arrange the letters of the word "MEDICAL" such that all the vowels are always together.
Quantity B: 700

A) Quantity A is greater B) Quantity B is greater
C) The two quantities are equal D) The relationship cannot be determined

190. In a pre-medical mock test, 60% of students passed in Biology, 70% passed in Chemistry, and 20% failed in both subjects. If 150 students passed in both subjects, what is the total number of students who took the test?

A) 300 B) 350 C) 400 D) 500

191. Let 'A @ B' mean 'A is the son of B'.
Let 'A # B' mean 'A is the brother of B'.
Let 'A * B' mean 'A is the mother of B'.
What does the expression 'X @ Y * Z # W' deduce about the relationship between X and W?

A) X is the uncle of W B) X is the brother of W C) W is the brother of X D) Y is the father of X

192. A man is facing North-West. He turns 90 degrees in the clockwise direction, then 135 degrees in the anticlockwise direction, and finally 180 degrees in the clockwise direction. Which direction is he facing now?

A) South-West B) South C) East D) North-East

193. Six friends P, Q, R, S, T, and U sit around a circular table facing the center.
- P sits second to the left of S.
- Q sits exactly opposite to S.
- T sits to the immediate right of S.
- U is not an immediate neighbor of Q.
Who sits exactly opposite to P?

A) R B) U C) T D) Q

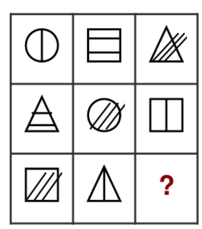
194. Five cars A, B, C, D, and E are parked in a row based on their weights.
Car A is heavier than Car C but lighter than Car D.
Car B is lighter than Car E.
Car C is heavier than Car E.
Which car is the lightest among all?

A) B B) C C) E D) Cannot be determined

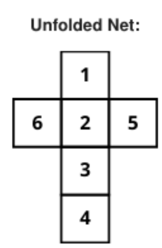
195. If February 29, 2024 fell on a Thursday, what day of the week will February 28, 2025 be?

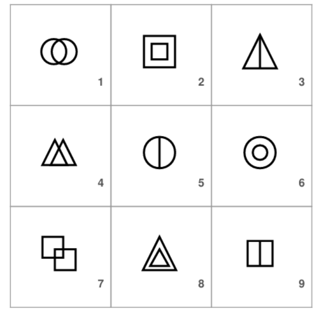
A) Wednesday B) Thursday C) Friday D) Saturday

196. Find the figure that completes matrix logic.



197. Cubes and Dice. Choose the box that can be formed by folding,
the given flat sheer (net) along the lines.



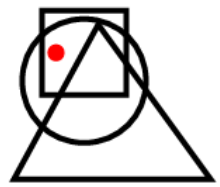


198. Group the given 9 figures into three distinct classes using each figure, exactly once based on their shared visual properties.

A. (1, 4, 7) ; (2, 6, 8) ; (3, 5, 9)
B. (1, 2, 3) ; (4, 5, 6) ; (7, 8, 9)

C. (1, 6, 7) ; (2, 4, 8) ; (3, 5, 9)
D. (1, 4, 7) ; (2, 5, 8) ; (3, 6, 9)

199. Dot Situation: In the problem figure, a dot is placed in a specific region constrained by overlapping geometric shapes. Select the option figure which contains a region satisfying the *exact same* set of conditions.



20. Find out which of the option figures hides the complex problem figure 'X' embedded inside it (rotation is NOT allowed).

Figure 'X